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Oefeningenreeks 5G nr. 1.11

$$f(x) = \tan x, x \in \left[0, \frac{\pi}{4}\right]$$

$$V = \pi \int_0^{\frac{\pi}{4}} \tan^2 x dx$$

$$= \pi \int_0^{\frac{\pi}{4}} (\sec^2 x - 1) dx$$

$$= \pi [\tan x - x]_0^{\frac{\pi}{4}} = \pi \left(\tan \frac{\pi}{4} - \frac{\pi}{4} \right)$$

$$= \pi \left(1 - \frac{\pi}{4} \right)$$

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References